

Reconfigurable DDR5 Memory Controller

MC Core — Design Specification

Sub-Blocks · FSMs · Timing Constraints · DFI 5.2 Interface

Standard	JEDEC JESD79-5, DFI 5.2, AXI4	Burst	BL16 (BL/2 = 8 clk)	FSMs	23 total
Banks	16 (2 BG × 4 banks)	Clock	Dual: AXI / MC (CDC at one crossing)	DFI ratio	1:1, 1:2, 1:4

1 System Overview

The MC Core sits entirely in the MC clock domain, receiving commands from the Client Interface (CIF) via a single Async FIFO CDC bar, and emitting DFI 5.2 commands to the DDR PHY. Every DRAM timing constraint, scheduling decision, refresh housekeeping, and power-state transition is the MC Core’s responsibility. It has no knowledge of the AXI protocol — that is fully resolved by the CIF upstream.

1.1 Full Microarchitecture Block Diagram

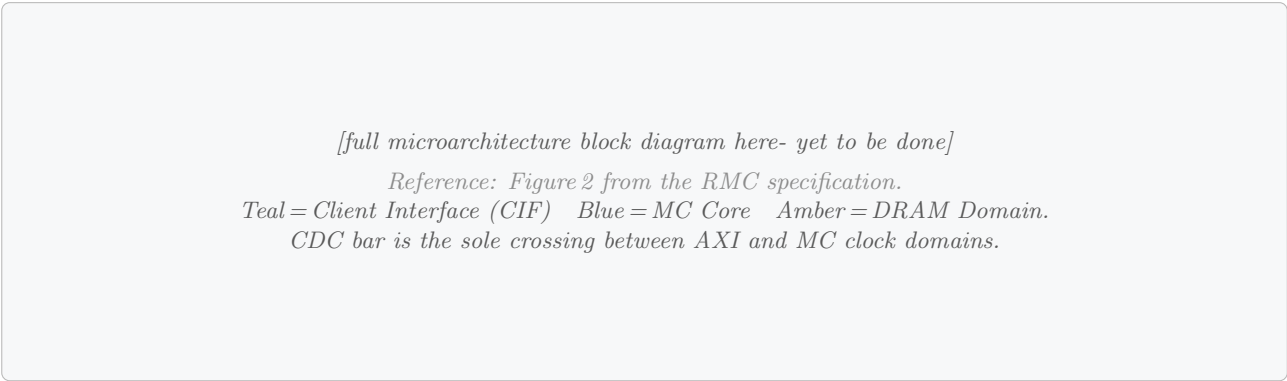


Figure 1. Full microarchitecture block diagram of the RMC.

1.2 Sub-Block Inventory

Sub-Block	Domain	Function
Config Registers	MC	All timing parameters, page policy, refresh mode, RFM thresholds, DFI off-sets. Written via AXI4-Lite CSR path.
Init FSM	MC	Full DDR5 power-up: RESET _n , NOP burst, DLL reset, ZQcal via MPC, MRW burst, optional training dispatch. Single instance.
Bank State Machines	MC	16 instances. Tracks bank state, owns all per-bank timers. Source of truth for bank availability.
Timing Control	MC	Combinational + registered timer array. Produces cmd_ok[15:0][cmd_type] each cycle for the Scheduler.
Arbitration Logic	MC	Picks winner command class each cycle. Priority, age-based anti-starvation, watermark RD/WR mode switching.
Command Selector	MC	Takes winner class, applies timing constraints, enforces page policy, emits {cmd, rank, BG, bank, row, col}.
Refresh Scheduler	MC	Leaky-bucket tREFI tracking. REFab and REFsb. Up to 8× postponement; ref_urgent on overflow.
ZQcal FSM	MC	Periodic MPC ZQCAL Start/Latch sequence.
RFM FSM	MC	DDR5 rowhammer mitigation. Per-bank RAA counters; triggers RFM when RAA ≤ RAAIMT.
Power Mgmt FSM	MC	CKE-controlled transitions: Active PD, Precharge PD, Self-Refresh, MPSM.
Write Data Path	MC	BL16-aligned write buffer, DM/ECC generator, Write CRC (CRC-8), DFI write timing pipeline.
Read Data Path	MC	Read capture FIFO, Read CRC/ECC checker, read latency counter, ECS FSM.

2 Config Registers

All timing parameters consumed by the MC Core live here. Programmed before INIT_KICK; may be updated at runtime only while the controller is idle. The Scheduler reads timing values combinatorially every cycle.

Register	Width	Description
CL	7b	CAS Latency (nCK). Drives RL in RTW formula.
CWL	7b	CAS Write Latency = CL−2 (fixed in DDR5).
tRCD	8b	RAS-to-CAS delay. ACT to first RD/WR on same bank.
tRP	8b	Row Precharge Time. PRE to next ACT on same bank.
tRAS	9b	Row Active Time minimum. ACT to PRE minimum.
tRC	9b	Row Cycle Time = tRAS+tRP. ACT-to-ACT same bank.
tWR	8b	Write Recovery Time. WR burst end to PRE.
tRTP	6b	Read-to-Precharge. RD cmd to PRE on same bank.
tCCD_L	5b	CAS-to-CAS, same BG (nCK).
tCCD_S	5b	CAS-to-CAS, diff BG = BL/2 = 8 nCK.
tCCD_L_WR	6b	WR-to-WR same BG. max(32 nCK, 20 ns).
tCCD_L_WR2	6b	WR-to-WR same BG, 2nd not RMW. max(16 nCK, 10 ns).
tWTR_L/tWTR_S	6b/5b	Write-to-Read same/diff BG.
tRRD_L/tRRD_S	5b/4b	ACT-to-ACT same/diff BG.
tFAW	6b	Four-Activate Window (nCK).
tRFC1	12b	REFab recovery time. 195 ns for 8 Gb DDR5.
tRFCsb	11b	REFsb per-bank recovery time.
tREFI	14b	Average refresh interval. 3.9 μs at ≤85°C.
tMRD	5b	MRW-to-MRW minimum. 16 nCK in DDR5.
tXP / tXS	5b/12b	PDX-to-command / SRX-to-command.
tCKSRE/tCKSRX	5b	Stable clock after/before SR entry/exit.
tDLLK	11b	DLL lock time. 1024 nCK.
RAAIMT/RAAMMT	8b	RFM RAA Initial/Maximum Management Threshold (from MR58).
RAADec	4b	RAA decrement per REF command issued.
REF_MODE	2b	00=REFab, 01=REFsb, 10=FGR-2×, 11=FGR-4×.
PAGE_POLICY	2b	00=Open, 01=Closed, 10=Adaptive.
AGE_THR1/2	8b	Intra/cross-class age escalation thresholds.
WR_HIGH/LOW_WM	6b	Write pool watermarks (default 32 / 8).
PHY_WRLAT	6b	dfi_t_phy_wrlat: WL pipeline offset to PHY.
PHY_RDLAT	6b	dfi_t_rddata_en: RL pipeline offset to PHY.
FREQ_RATIO	2b	DFI frequency ratio: 00=1:1, 01=1:2, 10=1:4.
INIT_KICK	1b	Write 1 to trigger Init FSM.
SOFT_RESET	1b	Synchronous soft-reset of all MC Core FSMs.

3 Init FSM

Reference: JESD79-5 §3.3.1/3.3.2. Single instance. Triggered by INIT_KICK. Emits DFI commands directly, bypassing the Cmd Scheduler. Asserts `init_done` on completion.

3.1 State Diagram

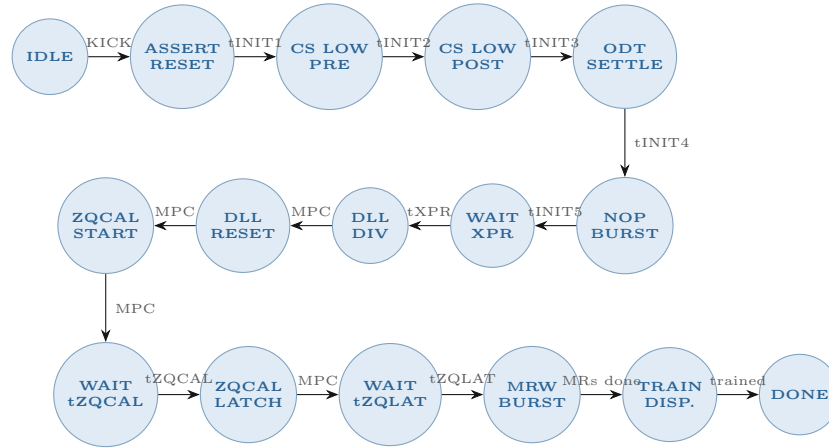


Figure 2. Init FSM state sequence (JESD79-5 §3.3.1). Executes once on INIT_KICK; idles until SOFT_RESET. DFI commands issued directly, bypassing Scheduler.

3.2 State Descriptions

State	Action	Exit
IDLE	Hold RESET_n=0, CKE=0, CS_n=0.	INIT_KICK=1
ASSERT_RESET	RESET_n LOW. CS_n LOW. Load tINIT1 (200 μ s).	tINIT1 expires
CS_LOW_PRE	CS_n LOW. Wait tINIT2 (10 ns) before releasing RESET_n.	tINIT2 expires
CS_LOW_POST	Deassert RESET_n. CS_n LOW. CKE LOW. Count tINIT3 (4 ms).	tINIT3 expires
ODT_SETTLE	Raise CKE. Stable CK running. Count tINIT4 (2 μ s).	tINIT4 expires
NOP_BURST	Issue ≥ 3 NOP/DES. CS_n driven normally.	tINIT5 satisfied
WAIT_XPR	Count $t_{XPR} = \max(5 \text{ nCK}, t_{RFC1} + 10 \text{ ns})$.	tXPR expires
MPC_DLL_DIVIDER	MPC: DLL Divider Select — configures tCCDL divider via MR13.	cmd accepted
MPC_DLL_RESET	MPC: DLL Reset. Required before ZQcal.	cmd accepted
ZQCAL_START	MPC: ZQCAL Start (OP=0x00).	cmd accepted
WAIT_tZQCAL	DRAM performs internal ZQ calibration.	1 μ s expires
ZQCAL_LATCH	MPC: ZQCAL Latch (OP=0x01).	cmd accepted
WAIT_tZQLAT	Wait tZQLAT = 30 nCK.	tZQLAT expires
MRW_BURST	MRW to MR0,2,4,5,6,8,32–35,58. One per tMRD (16 nCK).	all done
TRAINING_DISPATCH	CA/CS training, Write Leveling, Read training (TRAIN_EN).	training done
DONE	Assert init_done. Release Cmd Scheduler.	permanent

4 Bank State Machines (BSM)

16 instances, one per bank (2 BG $\times$ 4 banks $\times$ 2 sub-channels). Each BSM is independent, receives commands from the Cmd Scheduler, and owns all per-bank timers.

4.1 State Diagram

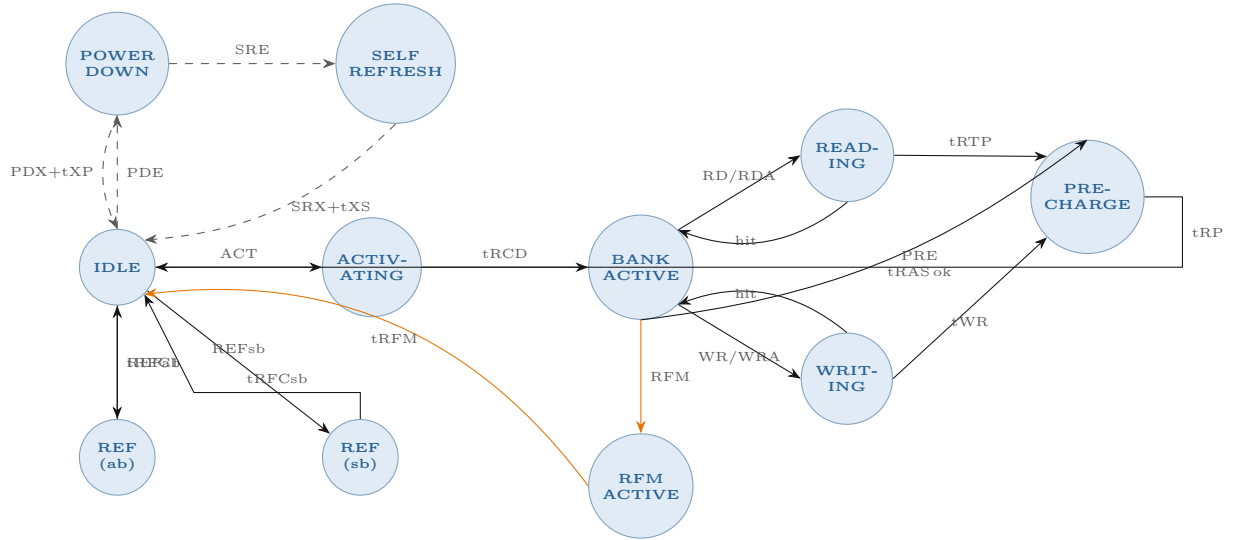


Figure 3. Bank State Machine (single bank, 16 instances). Dashed arrows: CKE-controlled power transitions. Orange arrows: DDR5 RFM path.

4.2 State Descriptions

State	Legal Inputs	Timer / Exit
IDLE	ACT, REFab, REFsb, PDE, SRE, RFM	No timer. Exits on received command.
ACTIVATING	(internal)	tRCD countdown → BANK_ACTIVE.
BANK_ACTIVE	RD, RDA, WR, WRA, PRE, PREA	tRAS loaded. PRE only after tRAS satisfied. Open row registered.
READING	BST	CL+BL/2. If RDA: hidden PRE starts at tRTP.
WRITING	BST	CWL+BL/2. If WRA: hidden PRE at CWL+BL/2.
PRECHARGING	—	tRP counter → IDLE.
REF (REFab)	—	tRFC1. All 16 banks locked simultaneously.
REF (REFsb)	—	tRFCsb. Only target bank locked; others accessible.
POWER DOWN	PDX	Precharge PD (from IDLE) or Active PD (from ACTIVE). Exits after tXP.
SELF REFRESH	SRX	tCKSRE after CKE falls. SRX: tCKSRX, then tXS.
RFM ACTIVE	—	tRFM (= tRFC1). Issued by RFM FSM when $RAA \leq RAAIMT$.

4.3 Per-Bank Timer Summary

Timer	Loaded On	Guards
tRCD	ACT issued	First RD/WR on same bank
tRAS	ACT issued	PRE/PREA on same bank (minimum open)
tRC	ACT issued	Next ACT on same bank
tRP	PRE issued	Next ACT on same bank
tWR	WR burst end	PRE on same bank
tRTP	RD issued	PRE on same bank
tRFC1	REFab issued	Any command on any bank
tRFCsb	REFsb issued	Commands to refreshed bank only
tXS	SRX asserted	First ACT after self-refresh exit
tXP	PDX asserted	First command after power-down exit
tMRD	MRW issued	Next MRW

5 Timing Control

Not a state machine. Combinational and registered timer array producing `cmd_ok[15:0][CMD_TYPES]` every MC clock cycle. The Scheduler may only issue a command when the corresponding `cmd_ok` bit is asserted.

5.1 Cross-Bank Constraint Checkers

Constraint	Scope	Implementation
tRRD.L	Same BG, any bank	Timestamp of last ACT in BG. Block until $T_{now} - T_{last_ACT_BG} \geq t_{RRD.L}$.
tRRD.S	Diff BG, any bank	Timestamp of last ACT globally. Block until $\Delta T \geq t_{RRD.S}$.
tFAW	All banks, same rank	4-entry ring buffer of ACT timestamps. Block if oldest is within tFAW.
tCCD.L	Same BG CAS	Last CAS in BG. Use tCCD.L.WR for WR; tCCD.L.WR2 for non-RMW WR.
tCCD.S	Diff BG CAS	Last CAS globally. $\Delta T \geq t_{CCD.S} = BL/2$.
tWTR.L	Same BG, WR→RD	Last WR data end in BG. Block RD until $\Delta T \geq WL + BL/2 + t_{WTR.L}$.
tWTR.S	Diff BG, WR→RD	Same using $t_{WTR.S}$.
tRTW	Any, RD→WR	Last RD cmd. Block WR until $\Delta T \geq RL + BL/2 - WL + 2 + t_{WPRE}$.

6 DDR Command Scheduler

Arbitration heart of the MC Core. Consumes `cmd_ok` from Timing Control, bank states from BSMs, and pending transactions from RD/WR Cmd Pools. Issues one DFI command per clock cycle.

6.1 Scheduler FSM

6.2 Command Priority Order

1. Refresh Override — REF budget about to expire (highest priority)
2. RFM — $RAA \leq RAA_{MT}$ (safety critical)
3. ZQcal — periodic; low urgency until overdue
4. Write drain — write buffer nearly full
5. Row-hit reads — open page, same row
6. Row-hit writes
7. Row-miss reads — ACT → wait tRCD → RD
8. Row-miss writes
9. PRE — close idle banks in closed-page mode

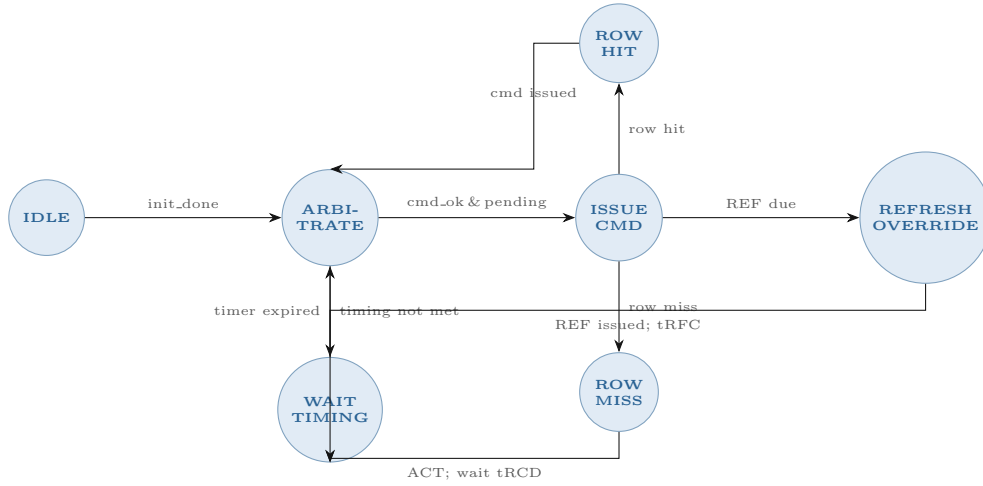


Figure 4. DDR Command Scheduler FSM. One command per MC clock cycle.

6.3 Page Policy

Policy	Row Hit	Row Miss	Best For
Open Page	CAS only	$\text{PRE} \xrightarrow{t_{RP}} \text{ACT} \xrightarrow{t_{RCD}} \text{CAS}$	Sequential / high locality
Closed Page	ACT+CAS	$\text{ACT} \xrightarrow{t_{RCD}} \text{CAS}$	Random / streaming
Adaptive	Hit-rate decides	Sliding window $N = 256$	Mixed; autotunes per bank

6.4 Arbitration Logic

Level	Winner	Condition	Effect
0	REF urgent	$\text{Credits} \geq 8$ or $\Delta t_{\text{sinceREF}} > t_{\text{REFI_max}}$	Hard stall; all RD/WR suppressed
1	REF nominal	tREFI countdown reaches zero	REF injected; RD/WR deferred
2	ZQ cal	tZQCS interval expires	ZQCS after current REF
3	Read	RD pool non-empty & RD mode	BG-aware scheduling
4	Write	WR pool non-empty & WR mode	Watermark-controlled batching
5	Scrubber/MRR	Background idle slot	Pre-emptible

Anti-starvation: age counter per entry. At AGE_THR1 (64 cy): intra-class HOL bypass. At AGE_THR2 (256 cy): WR may overtake RD priority. Resets to 0 on dispatch.

RD/WR mode switching: switch to WR mode when WR depth $\geq$ WR_HIGH_WM (32) or RD empty. Switch back when WR depth $\leq$ WR_LOW_WM (8) or WR empty or urgent RD. Hysteresis gap of 24 entries prevents mode thrashing.

6.5 Command Selector — Legal Check Matrix

All gates checked before dispatch. Any false bit $\Rightarrow$ NOP inserted that cycle.

Gate	Timer Source	Constraint
bank_avail[b]	BSM state	Bank not in ACTIVATING or PRECHARGING
gate_rcd[b]	ACT issue	t_{RCD} : ACT to first CAS on same bank
gate_rp[b]	PRE issue	t_{RP} : PRE to ACT on same bank
gate_ras[b]	ACT issue	t_{RAS} : ACT to PRE minimum
gate_ccd_l	Same BG CAS	t_{CCD_L} or $t_{CCD_L_WR}$ or $t_{CCD_L_WR2}$
gate_ccd_s	Diff BG CAS	$t_{CCD_S} = BL/2$
gate_rrd_l	Same BG ACT	t_{RRD_L}
gate_rrd_s	Diff BG ACT	t_{RRD_S}
gate_faw	Sliding window	≤ 4 ACTs within t_{FAW}
gate_wtr_l	WR data end, same BG	t_{WTR_L} : WR to RD
gate_wtr_s	WR data end, diff BG	t_{WTR_S} : WR to RD
gate_rtw	RD cmd	$t_{RTW} = RL + BL/2 + 2 - WL$
gate_rfc[r]	REF issue, per rank	t_{RFC1} : all commands blocked
gate_rfcpb[r] [b]	REFsb, per bank	t_{RFCsb} : same bank blocked
gate_zqcs	ZQCS issue	All commands blocked during ZQ
gate_mrd	MRW issue	t_{MRD} : MRW-to-MRW

7 Maintenance Engine

Four independent sub-FSMs handle all autonomous background operations. Each posts requests to the Cmd Scheduler via a priority request bus.

7.1 Refresh Scheduler FSM

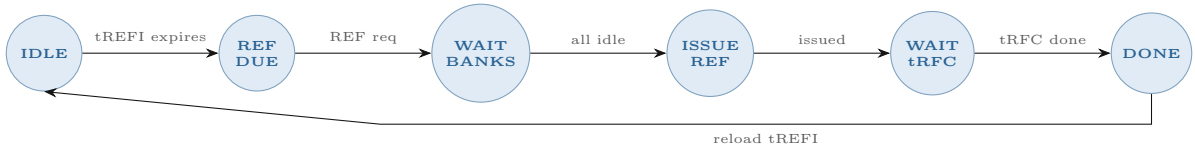


Figure 5. Refresh Scheduler FSM. REFsb path substitutes WAIT_TARGET.BANK for WAIT.BANKS and uses tRFCsb. FGR mode replaces tRFC1 with tRFC2/tRFC4.

- **Leaky-bucket:** +1 each tREFI, –1 on REF issued. Counter ≥ 9 : ref_urgent asserted.
- **REFsb cycling:** BA[1:0] cycles 00 $\rightarrow$ 01 $\rightarrow$ 10 $\rightarrow$ 11. 8 REFsb $\approx$ 1 REFab credit.
- **FGR:** REF_MODE[1]=1 $\Rightarrow$ tRFC2; REF_MODE=11 $\Rightarrow$ tRFC4. Budget threshold halved/quartered.
- **Temperature derating:** MR4 TUF=1 $\Rightarrow$ tREFI halved. MC polls MR4 via background MRR.

7.2 ZQcal FSM

States: IDLE $\rightarrow$ WAIT_IDLE $\rightarrow$ ISSUE_ZQCAL_START $\rightarrow$ WAIT_tZQCAL $\rightarrow$ ISSUE_ZQCAL_LATCH $\rightarrow$ WAIT_tZQLAT $\rightarrow$ DONE $\rightarrow$ IDLE.

Triggered by periodic tZQCS timer or host ZQCAL-TRIG bit. All banks must be idle. During ZQCAL, all commands blocked via gate_zqcs.

Parameter	Value	Description
t_{ZQCAL}	1 μ s	ZQCAL Start to Latch
t_{ZQLAT}	30 nCK	Latch to next command
t_{ZQCS_INT}	~ 128 ms	Periodic interval (CSR-configurable)

7.3 RFM FSM

States: IDLE $\rightarrow$ MONITOR_RAA $\rightarrow$ RFM.REQUEST $\rightarrow$ WAIT_ISSUE $\rightarrow$ WAIT_tRFM $\rightarrow$ UPDATE_RAA $\rightarrow$ MONITOR_RAA.

- RAA[b] += 1 on every ACT to bank b ; RAA[b] -= RAADec on every REF to bank b .
- RAA[b] $\leq$ RAAIMT: assert RFM request (priority above ZQcal, below refresh override).

- $t_{RFM} = t_{RFC1}$. Bank enters RFM_ACTIVE. After t_{RFM} : RAA[b] reset to RAAIMT.

7.4 Power Management FSM

States: NORMAL $\rightarrow$ PD_ENTRY_CHECK $\rightarrow$ {PRECHARGE_PD — ACTIVE_PD} $\rightarrow$ PDX_WAIT_ t_{XP} $\rightarrow$ NORMAL.

SR branch: NORMAL $\rightarrow$ SR_ENTRY $\rightarrow$ WAIT_ t_{CKSRE} $\rightarrow$ SELF_REFRESHING $\rightarrow$ SR_EXIT $\rightarrow$ WAIT_ t_{CKSRX} $\rightarrow$ WAIT_ t_{XS} $\rightarrow$ WAIT_ t_{DLLK} $\rightarrow$ NORMAL.

MPSM: MR2 OP[1:0]=01. Deeper than SR. Exit via DES then MRW to clear.

8 Write Data Path

Sub-Block	Description
Write Data Buffer	BL16-aligned FIFO. Stores write data from CIF Write Cmd Pool via CDC. Dequeued when Scheduler confirms a write slot.
DM/ECC Generator	Per-byte Data Mask. Inline Write CRC using $G(x) = x^4 + x + 1$ over DQ+DM bits per BL16 burst. Enabled via MR8 OP[6]=1.
WL Align	Programmable pipeline delay = CWL to align DQS preamble to dfi_wrddata_en. Offset from PHY_WRLAT.
DFI Write Timing	Drives dfi_wrddata[127:0], dfi_wrddata_mask[15:0], dfi_wrddata_cs_n, dfi_parity_in. Preamble per MR8 (2tCK). ODT: $t_{ODTon} = WL - 2$, $t_{ODToff} = WL + BL/2$.
Write CRC FSM	NORMAL $\rightarrow$ CRC_ERROR $\rightarrow$ RETRY_WRITE $\rightarrow$ AUTO_DISABLE. Monitors dfi_alert_n. Retries up to n times; auto-disables CRC via MRW to MR8.

9 Read Data Path

Sub-Block	Description
Read Latency Counter	Per-outstanding-read counter. Tracks CL + read preamble from RD command to expected DFI data arrival. Issues dfi_rddata_en PHY_RDLAT cycles early.
Read Data FIFO	Captures dfi_rddata[127:0] on dfi_rddata_valid. Re-aligns per-burst data. Feeds CIF Reorder Buffer with AXI transaction tag.
ECC/CRC Checker	Verifies Read CRC when MR8 OP[7]=1. On error: flags to CIF error register; optionally triggers read retry. Adds RL latency per JESD79-5 Table 365.
ECS FSM	IDLE $\rightarrow$ ECS_ACTIVE $\rightarrow$ WAIT_ t_{ECS} $\rightarrow$ IDLE. Triggered by MR14 or host MPC. DRAM scrubs on-die ECC during t_{ECS} ($\leq t_{RFC1}$). Logs in MR16–MR20.

10 DFI 5.2 Interface

dfi_wrddata_en asserted PHY_WRLAT cycles before write data. dfi_rddata_valid arrives PHY_RDLAT cycles after the RD command. Both offsets set during DFI training. FREQ_RATIO determines nCK per DFI clock.

Signal	Dir	Description
dfi_address[13:0]	MC→PHY	CA bus. 2-cycle encoding for DDR5 ACT.
dfi_cs_n[1:0]	MC→PHY	Chip select per rank.
dfi_cke[1:0]	MC→PHY	Clock Enable per rank.
dfi_odt[1:0]	MC→PHY	On-Die Termination control per rank.
dfi_reset_n	MC→PHY	DRAM RESET_n. Driven by Init FSM.
dfi_act_n	MC→PHY	Activate indicator (DDR5).
dfi_bank[1:0]	MC→PHY	Bank address.
dfi_bg[2:0]	MC→PHY	Bank group. BG[2:0] x4/x8; BG[1:0] x16.
dfi_wrdata[127:0]	MC→PHY	Write data (BL16 × DQ8).
dfi_wrdata_mask[15:0]	MC→PHY	Write data mask per byte lane.
dfi_wrdata_en	MC→PHY	Asserted PHY_WRLAT cycles early.
dfi_wrdata_cs_n	MC→PHY	Per-DRAM write chip select (DDR5 PDA).
dfi_parity_in	MC→PHY	CA parity bit (DDR5).
dfi_rddata[127:0]	PHY→MC	Captured read data.
dfi_rddata_valid	PHY→MC	Read data valid.
dfi_rddata_dnv	PHY→MC	Data-not-valid error.
dfi_alert_n	PHY→MC	DRAM alert: Write CRC or CA parity error.
dfi_ctrlupd_req/ack	both	Controller-initiated PHY update handshake.
dfi_phyupd_req/ack	both	PHY-initiated update/pause handshake.
dfi_init_start/complete	both	Init FSM PHY initialization handshake.
dfi_freq_ratio[1:0]	MC→PHY	00=1:1, 01=1:2, 10=1:4.

11 FSM Inventory

FSM	States	Inst.	Notes
Init FSM	16	1	RESET_n through training complete
Bank State Machine	11	16	One per bank; 176 total state bits
Arbitration Logic	6	1	RD/WR mode + priority selection
Command Selector	4	1	Page policy + legal check + seq gen
Refresh Scheduler	7	1	REFab + REFsb + FGR + urgent override
ZQcal FSM	7	1	MPC ZQCAL Start/Latch flow
RFM FSM	6	1	Per-bank RAA monitoring + RFM injection
Power Management FSM	10	1	PD (Pre/Act), SR, MPSM
Write CRC Error FSM	4	1	dfi_alert_n monitoring; retry + auto-disable
ECS FSM	4	1	DDR5 inline ECC scrub window
Total	≈75	23	

12 Timing Reference

Values at 200 MHz ($t_{CK} = 5\text{ ns}$, $CL = 3$, $CWL = 1$, $BL/2 = 8$) and 100 MHz. Production values set via Config Regs from actual speed bin.

Command Sequence	Formula	@200 MHz	@100 MHz
ACT → RD/WR (same bank)	tRCD	4	4
ACT → PRE (same bank, min)	tRAS	7	4
PRE → ACT (same bank)	tRP	4	4
ACT → ACT (same bank)	tRC = tRAS+tRP	11	8
ACT → ACT (same BG)	tRRD_L	4	4
ACT → ACT (diff BG)	tRRD_S	2	2
RD → RD (same BG)	tCCD_L	8	8
WR → WR (same BG)	tCCD_L_WR	32	32
WR → WR same BG (not RMW)	tCCD_L_WR2	16	16
RD/WR (diff BG)	tCCD_S = BL/2	8	8
RD → WR (any)	RL+BL/2-WL+2+tWPRE	14	14
WR → RD (same BG)	WL+BL/2+tWTR_L	13	13
WR → RD (diff BG)	WL+BL/2+tWTR_S	11	11
REFab → any	tRFC1	39	20
SRX → ACT	tXS = tRFC1+10 ns	41	21
ZQCAL Start → Latch	tZQCAL = 1 μ s	200	100
4 ACT window	tFAW	16	16

13 DDR5 vs DDR4 — Design Deltas

Feature	DDR4	DDR5 MC Core Impact
BL	BL8 (BL/2=4)	BL16 (BL/2=8). All RTW, WTR, WR-to-PRE formulae scale by 2.
ACT encoding	1-cycle CA	2-cycle CA. Scheduler drives CA on two consecutive rising edges.
REFsb	Not supported	Refresh Scheduler adds REFsb path; per-bank tRFCsb in BSM.
RFM	Not supported	Full RFM FSM; per-bank RAA counters; RAAIMT/RAAMMT in Config Regs.
ZQ mechanism	ZQCL/ZQCS cmds	MPC ZQCAL Start+Latch. Init FSM restructured; ZQcal FSM uses MPC.
tRFC1 (8 Gb)	350 ns	195 ns. Shorter penalty; more scheduling headroom.
tREFI	7.8 μ s	3.9 μ s. Double refresh rate; budget counter threshold halved.
CWL	Separate lookup	CWL = CL-2 fixed. Simplifies Config Reg and RTW formula.
tMOD	24 nCK after MRS	Replaced by tMRD only (16 nCK). No separate tMOD path.
tCCD_L_WR	Not defined	max(32 nCK, 20 ns). New gate in Command Selector.
Write CRC	Optional (MR5)	Mandatory support. dfi_alert_n monitoring; Write CRC FSM.
ECS	Not supported	ECS FSM in Read Data Path; hooks into tRFC window.
MR addressing	3 MRs	256 MRs via MRW/MRR. Init FSM MRW burst extended.

MC Core Design Specification — Memory Controller Capstone Project. JEDEC JESD79-5 (DDR5) / DFI5.2 / AXI4. Timing values device-specific; validate against datasheet per speed bin.